

Uses of the Masked Autoencoder Model in Later Literature

He et al. introduced masked autoencoders (MAE) as a scalable self-supervised vision-learning framework in which random image patches are removed, the encoder processes only the visible patches, and a lightweight decoder reconstructs the missing pixels from latent representations and mask tokens [1]. The central design choices—high masking ratios, asymmetric encoder-decoder computation, and reconstruction-based pretraining—have subsequently been adopted and modified across video, remote sensing, biomedical imaging, scientific imaging, multimodal learning, and non-image signal domains.

A major line of follow-on work extends the MAE paradigm from static images to video and spatiotemporal representation learning. BEVT applies masked image modeling first to image data and then combines masked image and masked video modeling for video transformers, using the image-pretrained transformer as a spatial prior for learning temporal dynamics [2]. AV-MaskEnhancer builds more directly on the reconstruction strategy of image and video masked autoencoders, arguing that visual-only MAE approaches can be limited when videos are low-resolution or blurry, and therefore adds audio-visual information to improve video representations [3]. Echo-Vision-FM adapts masked autoencoding to echocardiogram videos, explicitly using non-overlapping video patches, an asymmetric encoder-decoder architecture, and masked reconstruction during large-scale self-supervised pretraining before fine-tuning on downstream cardiac tasks [4]. Together, these studies show that the MAE model has served as a template for learning transferable spatiotemporal encoders under limited annotation.

A second group of papers applies MAE-style pretraining to remote sensing, Earth observation, and agricultural imagery. In remote sensing scene classification, masked-image-modeling-pretrained ViT backbones are evaluated on multiple scene classification datasets and reported to outperform supervised-learning counterparts [5]. In Earth observation, SatViT and related models are developed through masked autoencoding on remote-sensing data, with later extensions combining masked autoencoding and cross-modal contrastive learning for multimodal Earth-observation representation learning [6]. Agricultural image classification work similarly pretrains ViTs on a large agricultural image corpus using masked image modeling and then fine-tunes the pretrained model on agricultural classification datasets [7]. These applications preserve the core MAE premise that large unlabeled image corpora can be converted into useful supervised-task representations through masked reconstruction.

Medical and biological imaging also use the MAE model as a pretraining strategy for domains where annotation is costly. Histopathology work explores masked image modeling for whole-slide image representation learning and evaluates the resulting models across multiple cancer-related downstream tasks [8]. RamanMAE adapts masked autoencoders to Raman spectral data, using masked reconstruction to learn biologically meaningful spectral representations that support downstream molecular imaging tasks [9]. Echo-Vision-FM extends the same logic to echocardiogram video foundation modeling [4]. These studies indicate that MAE-like reconstruction objectives are especially attractive in biomedical settings because they exploit abundant unlabeled signals while reducing reliance on task-specific labels.

Several papers generalize the MAE architecture beyond conventional RGB imagery into scientific and physical measurement domains. A quaternion-based ViT-MAE applies masked autoencoding to polycrystalline EBSD orientation maps, using quaternion-valued representations and a 65% masking ratio to learn microstructural features from synthetic data that transfer to real scans [10]. PiMAE introduces a physics-informed masked autoencoder for optical microscopy, using self-supervised reconstruction to

estimate the point spread function and emitters directly from raw microscopy images [11]. A masked-autoencoder-based three-dimensional foundation model extends the approach to vortex identification by applying high masking ratios to PDE-derived flow-field data and fine-tuning the pretrained encoder for sparse-data vortex detection [12]. Optical hardware work implements MAEs using digital micromirror devices and multi-scale masked patches to reconstruct partially blocked scenes, linking the MAE reconstruction model to low-bandwidth optical sampling [13]. These papers treat MAE not simply as a computer-vision benchmark method, but as a general reconstruction-based representation learner for structured scientific signals.

The MAE idea has also moved into multimodal, neural, and behavioral data. scMMAE combines a masked autoencoder with cross-attention to fuse single-cell transcriptomic and proteomic modalities and to transfer multimodal knowledge to unimodal single-cell analysis [14]. An fNIRS foundation model uses a self-supervised masked autoencoder framework to pretrain on unlabeled fNIRS data and reduce the amount of labeled data needed for downstream classification [15]. SeqMAE adapts masked autoencoding to EEG-derived multiscale spatiotemporal features for brain activity decoding and downstream EEG-guided image generation [16]. hBehaveMAE applies masked autoencoders to motion-capture behavior data, using hierarchical latent representations to capture fine-grained movements at lower levels and more complex actions at higher levels [17]. Synthetic solar irradiance work also includes masked-autoencoder-based generative models for producing solar irradiance sequences used in forecasting and planning evaluations [18]. These examples show that later work often abstracts MAE away from images while retaining the defining mechanism of masked input reconstruction for transferable representation learning.

Finally, some later work uses MAE as a component in applied generative or hybrid systems. MAE-based image inpainting-steganography uses non-overlapping patches, random masking, an encoder-decoder reconstruction process, and feature-domain embedding to combine image inpainting with robust steganography [19]. In robotics, language-driven representation learning explicitly compares masked autoencoding with contrastive learning, reporting that masked autoencoding tends to capture low-level spatial features while contrastive learning captures higher-level semantic information; the proposed Voltron framework then combines language-conditioned visual reconstruction with visually grounded language generation [20]. This indicates a more mature phase of MAE use, in which later papers not only adopt the model but also analyze its representational strengths and limitations relative to contrastive and multimodal alternatives.

Overall, the confirmed literature suggests that the MAE model has become a reusable pretraining pattern: mask a substantial portion of structured input, learn to reconstruct missing content, discard or de-emphasize the decoder after pretraining, and transfer the encoder to downstream tasks. Later papers preserve this pattern while modifying the input type, masking strategy, reconstruction target, modality structure, or downstream objective. The strongest evidence of use appears in papers that explicitly name masked autoencoders, MAE, ViT-MAE, PiMAE, scMMAE, SeqMAE, hBehaveMAE, or masked auto-encoding frameworks.

Limitations

This review is based on the provided citation-classification prompt, including abstracts, classifier evidence, and extracted full-text snippets where available. It includes only papers classified as “yes” and with explicit evidence of MAE, masked autoencoder, masked auto-encoding, or closely related masked image modeling use. It does not discuss papers classified as uncertain or no-use, and it also excludes weak “yes” cases

where the available abstract evidence only mentioned generic pretraining, representation learning, transformers, or benchmarking without clear use of the MAE model.

References

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